A nonuniform witness for Jensen–Toft 15.6

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Abstract

We construct an eleven-vertex hypergraph with twenty edges of size four and four edges of size five. It is three-chromatic, and its weight $\sum_e 2^{-|e|} = 11/8$ is below $23/16$, the minimum weight of a four-uniform hypergraph without Property B. Thus the uniform-attainment question in Jensen and Toft's Problem 15.6 has a negative answer.

1 The question and the construction

A proper coloring of a hypergraph has no monochromatic edge. For $c > 1$, put

$$w_c(G) = \sum_{e \in E(G)} c^{-|e|}, \qquad f(n) = \inf\{w_2(G) : \chi(G) = 3,\ |e| \geq n \text{ for all } e \in E(G)\}.$$

Let $m(n)$ be the smallest number of edges in an n-uniform hypergraph without Property B. Jensen and Toft [1, Problem 15.6, pp. 235–236] ask whether $f(n)$ is attained by an n-uniform three-chromatic hypergraph. Since $m(4) = 23$ [2], every four-uniform three-chromatic hypergraph has weight at least $23/16$. Cherkashin [3, §2] gave a nonuniform example of weight $95/64$, which remains above this benchmark.

Theorem 1. *There is a three-chromatic hypergraph H with twenty four-edges and four five-edges such that*

$$f(4) \leq w_2(H) = \frac{11}{8} < \frac{23}{16} = \frac{m(4)}{2^4}.$$

Consequently no four-uniform hypergraph attains $f(4)$.

Proof. Take four disjoint pairs

$$A = \{a_0, a_1\}, \quad B = \{b_0, b_1\}, \quad C = \{c_0, c_1\}, \quad D = \{d_0, d_1\},$$

and a disjoint three-element set U. Write $T_{ijk} = \{a_i, b_j, c_k\}$ for $(i, j, k) \in \{0, 1\}^3$. The edges of H are exactly

$$P \cup S \quad (P \in \{A, B, C, D\},\ S \in \tbinom{U}{2}), \tag{I}$$

$$T_{ijk} \cup \{d_\delta\} \quad (i + j + k \equiv 0 \pmod 2,\ \delta \in \{0, 1\}), \tag{II}$$

$$T_{ijk} \cup D \quad (i + j + k \equiv 1 \pmod 2). \tag{III}$$

These families have twelve, eight, and four edges, respectively, and are disjoint. Thus H has eleven vertices, twenty four-edges, and four five-edges.

Suppose that a proper red–blue coloring exists. Interchange the colors if necessary so that at least two vertices of U are red. By (I), each of A, B, C, D contains a blue vertex. If one of A, B, C were wholly blue, choose blue vertices in the other two pairs and then choose from the wholly blue pair the vertex that makes the index sum even. A blue vertex of D would complete a blue edge of (II). Hence each of A, B, C has one blue and one red vertex.

The blue transversal of A, B, C must have odd index sum, or a blue vertex of D would complete an edge of (II). Its complementary red transversal has even index sum, because all three indices change. A red vertex of D would therefore complete a red edge of (II). Both vertices of D are consequently blue. The odd blue transversal together with D is now a blue edge of (III), a contradiction.

For a proper three-coloring, color A, C, D with color 0, B with color 1, and U with color 2. Each edge of (I) meets U and a pair; each edge of (II) or (III) meets both A and B. Thus $\chi(H) = 3$. Finally,

$$w_2(H) = 20 \cdot 2^{-4} + 4 \cdot 2^{-5} = \frac{11}{8} = \frac{22}{16} < \frac{23}{16}.$$

The four-uniform lower bound stated above proves the claim. □

Theorem 2. *For every $4/3 < c < 2$, the infimum of the c-weights among three-chromatic hypergraphs with edge sizes at least four is strictly less than the weight of every four-uniform three-chromatic hypergraph.*

Proof. The same H has

$$w_c(H) = 20c^{-4} + 4c^{-5} < 23c^{-4} \quad \Longleftrightarrow \quad c > \frac{4}{3}.$$

Every four-uniform three-chromatic hypergraph has at least 23 edges. □

The construction provides an upper bound for $f(4)$; it does not determine its exact value. The finite encoding and verification files are prepared for the companion repository at https://github.com/alexgocardinal/jt156-uniform-attainment. The repository link becomes active when the files are posted there.

References

[1] T. R. Jensen and B. Toft, *Graph Coloring Problems*, Wiley, New York, 1995, Problem 15.6, pp. 235–236.

[2] P. R. J. Östergård, On the minimum size of 4-uniform hypergraphs without property B, *Discrete Appl. Math.* **163** (2014), no. 2, 199–204. https://doi.org/10.1016/j.dam.2011.11.035.

[3] D. Cherkashin, On small non-uniform hypergraphs without property B, *Mathematics and Education in Mathematics* **52** (2023), 71–75. https://doi.org/10.55630/mem.2023.52.71-75.